

The University of Texas at Austin
IN-CLASS WORK 4

M339G Predictive Analytics

September 30, 2026

K-NEAREST NEIGHBORS.

Problem 4.1. (10 points) Explain why standardization is an appropriate preliminary step before we employ the K –nearest neighbors procedure.

Solution. Solutions will vary. The salient point of any response which is to earn credit must be that K –nearest neighbors employs a distance which is heavily affected by the natural scale of various predictors.

Problem 4.2. (10 points) You are using K –nearest neighbors in a classification problem with X as the explanatory variable and Y as the response. Here are the observed values:

x	1	2	5.5	6.5	9	13
y	1	2	1	1	2	2

Using $K = 3$, figure out the misclassification error.

Solution. With the neighbourhood of size 3, we get the following predictions:

x	1	2	5.5	6.5	9	13
$\hat{y}$	1	1	1	1	1	2

We did fine: the misclassification error rate is $1/3$.

Problem 4.3. (10 points) You are using K –nearest neighbors in a classification problem with $X = (X_1, X_2)$ as predictors and Y as the response. Here are the observed values:

x_1	1	1	2	2
x_2	1	4	2	-1
y	1	2	1	2

Using $K = 3$, figure out how the above points would be classified and the misclassification error. Then, state how you would classify point $(2, 4)$.

Hint: Draw a picture in the plane of (x_{i1}, x_{i2}) for $i = 1, 2, 3, 4$.

Solution. With the neighbourhood of size 3, we get the following predictions:

x_1	1	1	2	2
x_2	1	4	2	-1
y	1	1	1	1

We did fine: the misclassification error rate is $1/2$. The new point would be classified as 1.